

NIRANJAN KHOTRA			E-Mail : niranjankhotra@gmail.com	
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PROFILE SUMMARY				
Engineering graduate from IIT Hyderabad having passion in data science, with hands-on experience in demand forecasting, operational analytics and workforce optimization utilizing expertise in data science.				
EDUCATIONAL QUALIFICATIONS				
Degree		Institute	Year	Marks
M.Tech Electrical Engineering		Indian Institute of Technology, Hyderabad	2025	8.18 CGPA
B.Tech Electrical Engineering		Sardar Patel College of Engineering, Mumbai	2022	7.48 CGPA
Class XII (Maharashtra Board)		MH Junior College, Thane, Maharashtra	2016	71.54 %
Class X (Maharashtra Board)		Saraswati Secondary School, Thane, Maharashtra	2014	92.00 %
ACADEMIC ACHIEVEMENTS				
▪ CAT 2023 : Received admission offer for the flagship Post Graduate Programme in Management (PGP) at IIM Indore. ▪ GATE 2023 : Achieved a national rank of 4554 out of 55,292 candidates, placing in the top 8.24% of the examination.				
PROFESSIONAL EXPERIENCE				
Tata Power Company Ltd, Mumbai		Lead Engineer (Grid Operations)	Jun 2022 – Jul 2023	
▪ Managed real-time monitoring & control of Mumbai's electricity distribution network, ensuring proper demand-supply balance. ▪ Coordinated with grid operators and field engineers for translating complex technical data into actionable insights. ▪ Developed data-driven models to forecast electricity demand of Mumbai city using regression & optimization techniques. ▪ Designed an advanced dashboard to monitor the operational status of prime and essential consumers in real time mode. ▪ Constructed a geographical fault density map to optimize the workforce placement & resource allocation. ▪ Built real-time dashboards in Tableau and Power BI to track supply, demand (actual vs forecasted) and purchase prices. ▪ Automated daily & monthly report making process using Python and Excel macros, reducing manual effort and human errors. ▪ Achieved highest possible rating (P5) in the rigorous annual appraisal process for showcasing the outstanding performance.				
TECHNICAL SKILLS & TOOLS				
Python (Pandas, NumPy, Matplotlib, Scikit-learn) SQL Microsoft Excel (Pivot Tables, Macros, VBA) Power BI Tableau				
RELEVANT PROJECTS				
Demand Forecasting for Inventory Optimization Using Machine Learning				
▪ Built a demand forecasting model using Linear Regression and XGBoost to predict inventory needs and optimize stock levels. ▪ Extracted temporal features (year, month, holiday) and applied seasonality encoding transformations for better trend detection. ▪ Conducted EDA using Seaborn & Matplotlib to identify sales trends, seasonal variations and correlations affecting demand. ▪ Used Simple Moving Average (SMA) analysis to track long-term sales trends and improve stock replenishment decisions. ▪ Evaluated multiple regression models, with XGBoost achieving MAE score of 6.92, ensuring effective demand prediction. ▪ Designed a data-driven inventory allocation strategy, improving stock availability and reducing instances of overstock / stockout.				
Store Location Optimization Using Machine Learning				
▪ Developed a location selection model using K-Means clustering to identify high-demand areas for new store placement. ▪ Applied geo-spatial analysis to map ideal store locations, leveraging GIS tools to visualize demand concentration. ▪ Built a predictive model to estimate store profitability using historical sales, income levels and local demand trends.				
Customer Segmentation Using K-Means Clustering				
▪ Implemented customer segmentation using K-Means, optimizing clusters to identify distinct consumer behaviour patterns. ▪ Performed exploratory data analysis using Seaborn and Matplotlib to visualize customer demographics and spending habits. ▪ Created heatmaps to analyse feature correlations, identifying key attributes that influence customer purchasing behaviour. ▪ Extracted temporal features from the dataset, including customer tenure and seasonal spending trends to enhance the model.				
Digital Twin for Power Transformer				
▪ Applied data driven ML models for predictive maintenance and fault diagnostics, improving reliability of a transformer. ▪ Developed virtual sensors using regression models to estimate high-voltage parameters from low-voltage side data.				
POSITIONS OF RESPONSIBILITY				
Student Coordinator between Center for Continuing Education, SPCE and SITRAIN, Siemens :				
▪ Under the Technical value added courses in collaboration with industry, coordinated 10 batches consisting of 195 students (UG & PG) of Electrical and Mechanical department, which turned out to be the largest beneficiary program of CCE.				
EXTRA-CURRICULAR ACTIVITIES				
Volunteering Activities for Sustainable Development (In collaboration with SPJIMR)				
▪ Provided voluntary teaching to underprivileged scholarship students, contributing to UNSDG-4 (Quality Education). ▪ Visited various slums of Mumbai to conduct no-plastic awareness program, contributing to UNSDG-13 (Climate Action). ▪ Engaged in a discussion on Rural Development with Mr. Himanshu Patel, Sarpanch of India's first smart village, Punsari.				
Key Roles and Achievements in Motorsport Engineering Competitions for Team SPCE Racing				
▪ Served as Head of the Electrical Department for Formula Bharat 2020, securing 6th place overall among 70 Teams ▪ Established the 'Data Acquisition & Analysis' department to analyse vehicle behaviour and optimize the performance.				